The Functional Role of PING — a Measured Corpus

ar016 · 12 July 2026

This article is a **corpus**, not a review. It is the exhaustive, auditable paper list for one narrow question — the map — with the synthesis, the open debates, and the gaps left for a later pass. The question: **in which papers is the PING mechanism explicitly the vehicle for a functional claim?** PING here means gamma generated by the reciprocal pyramidal↔interneuron loop with its characteristic excitatory–inhibitory timing; a paper is in the corpus only if it both invokes that mechanism as the generator of the rhythm **and** argues that a cognitive or computational function follows from it — attention, stimulus selection, communication-through-coherence, gain control and coincidence windows, phase coding, or binding. Papers where gamma merely **correlates** with a function, or where the mechanism is studied with no functional claim, are out. The scope was frozen and tagged (`freeze-ar016-v1`) before any measured run, so the recall number below is valid — both samples target the same fixed population.

Confidence header

Papers in the corpus	161, every DOI verified
Estimated true size	≈ 180–250 (two-source Chapman 246, 95% CI 175–316)
Estimated recall	≈ 65–90 %
Channels	memory · keyword-search · citation-graph · embedding
Membership authority	one two-stage judge (wide-net → confirm), consistent across channels
Coverage caveats	keyless (no Semantic Scholar / PubMed); single judge per paper; OpenAlex abstract gaps on seminal papers

The estimate is a **range, not a point**, and that is deliberate. The three pairwise capture–recapture estimates disagree — 246, 219, and 142 — and that disagreement is diagnostic, not noise: the channels have heterogeneous catchability (memory reaches the old and seminal, search the recent and well-indexed), which violates the independence the estimator assumes. No single $\hat{N}$ is trustworthy, so the honest output is the band 180–250 and a recall of 65–90 %.

Method — four decorrelated channels, one judge

The premise is a division of labour. The model is a good **size oracle** and a good **membership judge** but a bad **recall oracle** — it can tell hundreds from thousands, and can judge whether a specific paper is in scope, but it cannot list the hundreds without fabricating. So recall is supplied by databases and the citation graph, membership is decided by judgment, and every candidate DOI is verified against Crossref before it is allowed in — an unresolvable DOI is treated as a hallucination and discarded.

Recall was gathered from four channels chosen to be **orthogonal** — to fail on different papers, so their union covers what any one misses:

- **Memory** — 20 decorrelated elicitation passes (per-lab, per-function-axis, per-decade), each in its own context, unioned. Owns the old, seminal, abstract-less stratum.
- **Keyword search** — a deep OpenAlex full-text sweep across the frozen synonym ring, deliberately high-recall. Owns the recent, well-indexed stratum.

- **Citation graph** — the backward references and forward citers of the found set, scored by how many of the corpus each candidate connects to. Owns the well-connected stratum, independent of words and of memory.
- **Embedding** — SPECTER nearest-neighbours (Semantic Scholar) of the strongest seeds, independent of shared words, citations, and authorship.

Membership was **not** decided by any keyword filter — an earlier attempt to do so failed, because a filter strict enough to exclude the broad "gamma and attention" literature also excludes the function-forward canonical papers whose abstracts never say "interneuron". Instead every candidate from every channel was passed through one consistent two-stage judge — a wide-net first pass, then a strict confirming pass against the frozen rubric — so the in/out boundary is one authority applied uniformly. Completeness was then estimated by Lincoln–Petersen / Chapman capture–recapture over that single judged population, treating each channel as an independent capture.

Results — funnel, provenance, saturation

The four channels surfaced 3111 distinct candidate papers, every one put through the same membership judge. 161 survived; 2950 were rejected — an acceptance rate of 5.2 %. That selectivity is the scope doing its work: most candidates are gamma-and-cognition papers that never make the mechanism-**carries**-function argument, and the judge holds the line.

Candidates judged (search + memory union)	1694
Candidates judged (citation graph)	1154
Candidates judged (embedding)	263
Total distinct papers judged	3111
Accepted into the corpus	161 (5.2 %)
Rejected	2950

The corpus is genuinely the union of complementary channels; no single one would have built it:

Found by memory only	59
Found by search only	45
Found by memory and search	21
Found by citation graph only	36
Total	161

Only 21 of the 161 papers were caught by both memory and keyword search — the low overlap is real channel complementarity, and it is **why** it took four channels rather than one.

The stopping decision rests on the marginal-yield curve, not on hitting a target. Each successive channel returned fewer new in-scope papers:

Channel	**New in-scope**	**Candidates screened**
memory	80	—
keyword search	+45	1552
citation graph	+36	1154
embedding (SPECTER)	+0	263

The yield collapsed from 80 to +45 to +36 to +0. The final channel — orthogonal, abstract-rich, centred on the hundred strongest seeds — screened 263 semantic neighbours and added nothing. That is the saturation signal: not a proof (nothing here is), but the convergent kind — falling returns together with an orthogonal null. The result is discounted only slightly, because SPECTER's recommendations are recency-biased and so its null speaks most strongly for the recent stratum; but the old stratum was already swept by the citation channel, so between the two the un-caught in-scope region looks small.

Discussion — what the number means and does not

What can be claimed: this is a DOI-verified corpus of 161 papers, with a **measured** — not assumed — recall of roughly 65–90 %, and a documented reason for every paper's inclusion. What cannot: that it is complete. The tail of a semantic corpus goes asymptotic — papers in no index, mis-tagged, or truly obscure cost more per paper than the bulk did, and the capture–recapture band (180–250) is itself uncertain because the channels are heterogeneous. A notable structural fact surfaced along the way: the most-cited **references** of this corpus are foundational papers that are purely mechanism (how gamma is generated) or purely function (that attention modulates gamma) — each satisfies only half the "mechanism **carries** function" test, which is why the citation graph flooded us with candidates but converted few. The scope boundary is genuinely restrictive; the intersection is the interesting, and smaller, set.

Next steps

The corpus is the input to the review, not the review. The obvious continuations: (1) a second, independent judge pass to turn the single-judge boundary into a voted one and tighten precision; (2) one more orthogonal recall channel where a key exists — PubMed for the biomedical silo, or author-complete feeds for the core labs — expected, from the yield curve, to add only single digits; (3) the synthesis itself — grouping the 161 papers by function axis, mapping which mechanistic claims are contested, and naming the gaps. Scope reopens only under a new freeze tag; questions that surfaced during the build are logged, not silently acted on.

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